Frequency response. Nyquist plot and stability

R0005E - Measurement and Feedback Control

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Nyquist plot

We assume that we have derived the frequency response $H(j\omega)$ of the transfer function $H(s)$. Then we can rewrite $H(j\omega)$ into the following form:

$$H(j\omega) = Re\left\{H(j\omega)\right\} + jIm\left\{H(j\omega)\right\}$$

When we have evaluated the real and imaginary part for the frequency range $\omega = -\infty \ldots \infty$, and plot the resulting line in the complex plane, then we get the Nyquist plot of the transfer function.

Revisit of the example:

Let's assume we want to determine the Nyquist plot of the following system:

$$\dot{y} + \frac{1}{40}y = \frac{5}{80}u \tag{1}$$

First we derive the transfer function for (1):

$$H(s) = \frac{2.5}{40s+1}$$

The frequency response is now given by

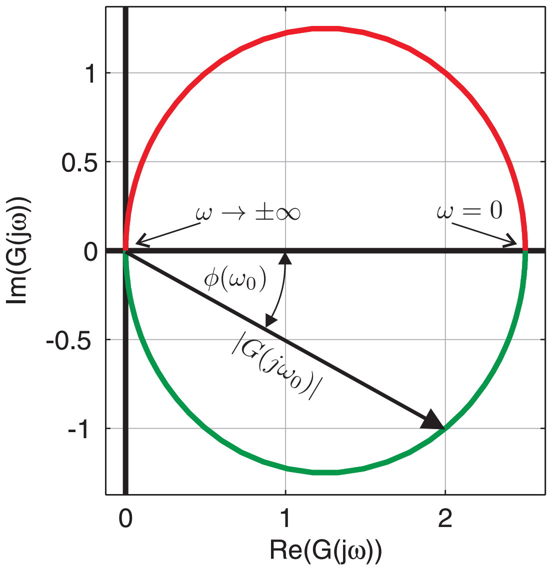
$$H(j\omega) = \frac{2.5}{40j\omega+1}$$

Now we can rewrite the frequency response into a real and imaginary part:

$$H(j\omega) = \frac{2.5}{40j\omega+1} \cdot \frac{-40j\omega+1}{-40j\omega+1} = \frac{2.5-100j\omega}{1600\omega^2+1} = \frac{2.5}{1600\omega^2+1} + j\frac{-100\omega}{1600\omega^2+1}$$

Nyquist plot

Now we can plot the line into the complex plane.



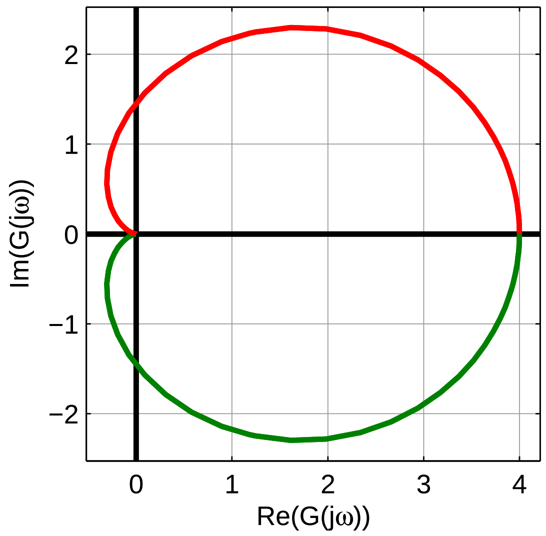
There: $\omega < 0$ (red), $\omega > 0$ (green).

Notes:

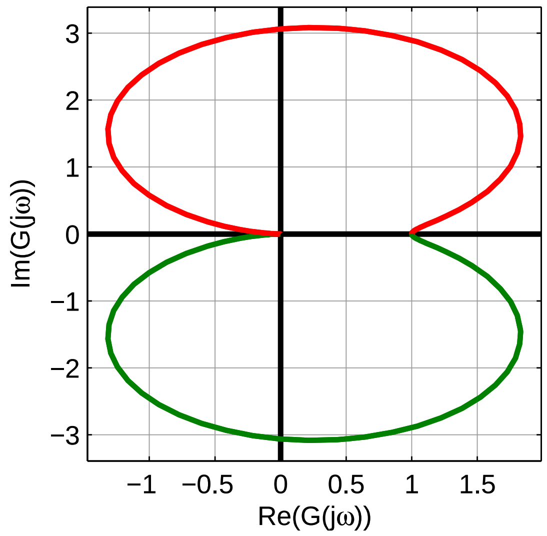
- It can be shown that $H(j\omega) = H^*(-j\omega)$, which means that the branch for the negative frequencies is the branch for the positive frequencies, but mirrored at the real axis.
- If the denominator has higher order than the numerator, then the plot ends in origo as $\omega \to \infty$.
- The branch for the positive frequencies and the bode plot can be directly converted into each other.

More examples:

$G(s) = \frac{4}{26s^2+14s+1}$



$G(s) = \frac{1.5}{s^2+0.4s+1.5}$



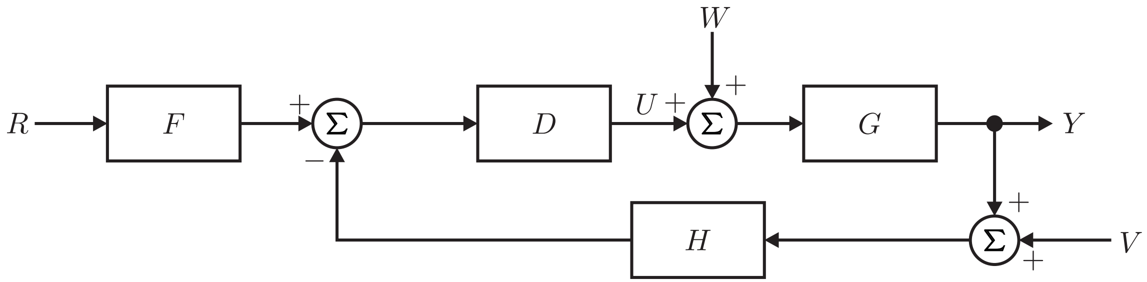
More information on Nyquist:

- Complex analysis
- Argument principle

Question: How can we now analyze stability using a Nyquist plot?

General stability condition

Assume that we have the following closed loop system:



where $H = 1$, $F = 1$, $W = 0$ and $V = 0$. We also denote $L = GD$ as the open loop transfer function.

Stability: The roots of $1 + L(s)$ have to be in the left half plane.

Argument Principle: A contour map of a complex function will encircle the origin $N = Z - P$ times, where Z is the number of zeros and P is the number of poles of the function inside the contour.

In this case:

$$1 + L(s) = 1 + \frac{b(s)}{a(s)} = \frac{a(s) + b(s)}{a(s)}$$

The poles of $L(s)$ are also the poles of the function and $a(s) + b(s)$ gives rise to the zeros of the function.

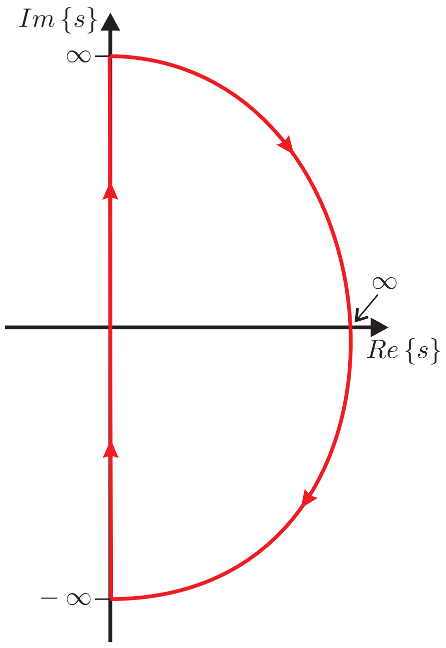
Idea: Choose as contour the complete RHP and then the encirclements of the origin gives us the number of zeros.

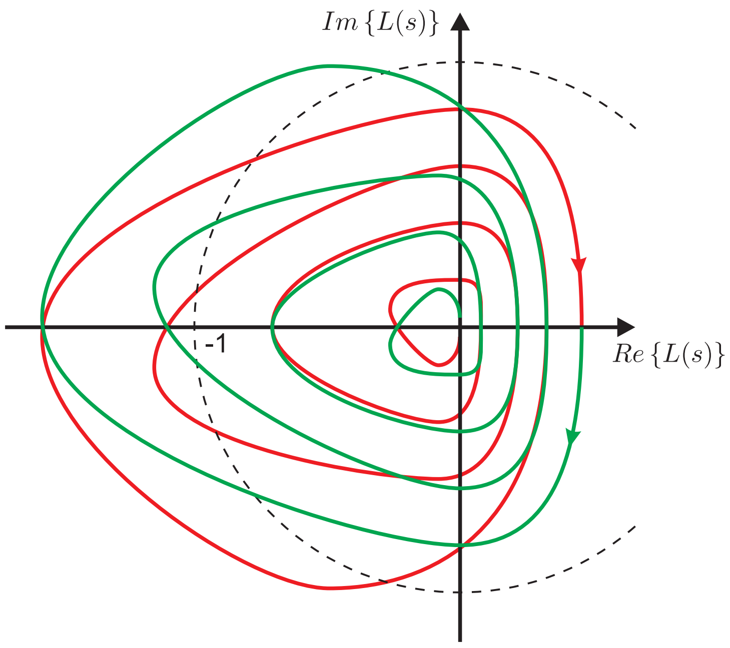
General stability condition (2)

But: We want to assess the open-loop transfer function to get information on the closed loop.

Evaluation of the complex function $1 + L(s)$ is the same as evaluating the complex function $L(s)$ and shifting it to the right by 1.

Consequence: We can evaluate $L(s)$ and assess the encirclement of -1. Then we have evaluated the encirclements of the origin by $1 + L(s)$.





Procedure:

1. Evaluate $L(s)$ for $-j\infty \ldots j\infty$ and plot the result in the complex plane. In Matlab use the command `nyquist`.

2. Derive the number of clock-wise encirclements of -1 and denote it N. This can be done drawing a straight ray out of -1. Left to right crossings are counted positive and right to left a counted negative.

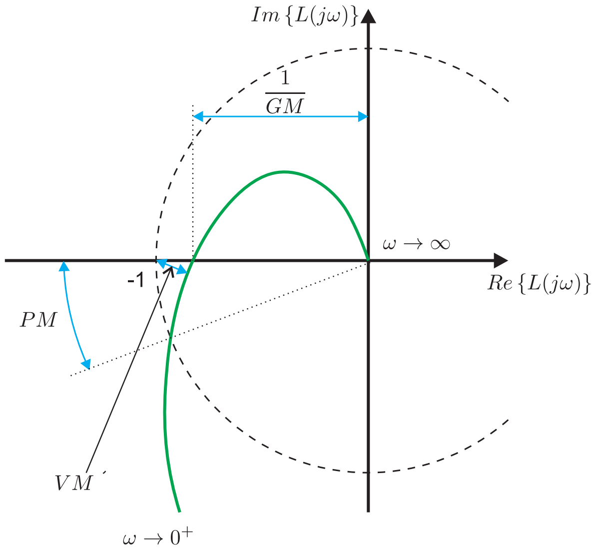
3. Determine the number of unstable poles of $L(s)$ and denote the number P.

4. Calculate the number of unstable poles of the closed loop system (zeros of the characteristic equation) Z by
$$Z = N + P$$

Stability margins

Similarly, to the bode plot the stability margins can be defined in a nyquist plot:

- **Phase margin:**
  - Definition: Angular distance from -180° when $|L(j\omega)| = 1$.
  - Interpretation: This is the point where the nyquist plot enters the unit circle around the origin. This point is also called cross-over.
- **Gain margin:**
  - Definition: Inverse of the magnitude when $\angle L(j\omega) = -180^\circ$.
  - Interpretation: This is the point where the nyquist plot crosses the negative real axis.



Vector margin

The vector margin is more accurate in quantifying the distance to the stability border. This is especially interesting, when the process model is not accurate, i.e. model uncertainties.

We know:

- Stability is evaluated using the complex function $1 + L(s)$
- A complex number can be interpreted as a vector.

Consequence: $1 + L(s)$ is actually the distance between -1 and $L(s)$.

Remember: The sensitivity function S was defined as the ratio between relative change in closed loop transfer function due to a relative model error.

$$S = \frac{1}{1 + L(s)}$$

Now we can analyze the frequency response of S. Obviously, the magnitude of $S(j\omega)$ is the inverse of the distance to the point -1.

Consequence: The closest point on $L(j\omega)$ to -1 is the maximum value of $|S(j\omega)|$. A large peak in $|S(j\omega)|$ should therefore be avoided.

More on that: R7014E - Advanced Control Systems.

Bode's gain-phase relationsship

Bode's theorem: For any stable minimum-phase system $G(s)$, the phase of $G(j\omega)$ is uniquely related to the magnitude of $G(j\omega)$

Two cases:

1. If the slope of the magnitude plot is a constant n, then the phase will be

$$\angle G(j\omega) \cong n \cdot 90^\circ$$

2. If the slope is varying then

$$\angle G(j\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{dM}{du} W(u) du$$

with

$$M = \ln |G(j\omega)|$$
$$u = \ln \frac{\omega}{\omega_0}$$
$$W(u) = \ln \left\{ \frac{coth|u|}{2} \right\}$$

Note:

- Can be used to check the correctness of a bode plot.
- Limits the performance of the closed loop system.

Additional material

There are two videos that explain the usage of the Nyquist criterion

- Nyquist Stability Criterion, Part 1
  `https://www.youtube.com/watch?v=sof3meN96MA`
- Nyquist Stability Criterion, Part 2
  `https://www.youtube.com/watch?v=tsg0stfoNhk`